Dedekind-Kummer Index Theorem

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Conventions

"Ring" always means commutative ring with unity, ring homomomorphisms are assumed to send 1 to 1, a "prime" of a domain means a nonzero prime ideal. A "number ring" is the ring of integers in a number field.

1 Introduction

Fix number fields $K \subset L$ and their corresponding rings of integers $\mathcal{O}_K \subset \mathcal{O}_L$. A fundamental question in Algebraic Number Theory is the following: given a prime $\mathfrak{p} \subset \mathcal{O}_K$, determine the factorization of $\mathfrak{p}\mathcal{O}_L$. In many cases, in fact almost every, the Dedekind-Kummer Index Theorem (DK theorem for short) gives an explicit answer.

Recall that domain R is a **Dedekind Domain** provided it satisfies

1. R is Noetherian
2. R has Krull dimension one. That is, every nonzero prime ideal of R is maximal.
3. R is integrally closed in its field of fractions. That is, for all $\theta \in \mathrm{Frac}(R)$ we have $\theta \in R$ if and only if there is a monic $f \in R[x]$ such that $f(\theta) = 0$.

In particular all number rings are Dedekind domains. It is this class of Dedekind domains we will study today, but the DK theorem can be stated for general extensions $R \subset S$ of Dedekind domains.
Number rings have many subrings, but those subrings need not be integrally closed. For example the subring $\mathbb{Z}[\sqrt{5}] \subset \mathbb{Z}\left[\frac{1+\sqrt{5}}{2}\right]$ is a Krull dimension one Noetherian ring, but is not integrally closed in its field of fractions $\mathbb{Q}(\sqrt{5})$. Even though for every field extension $K \subset L$ we can write $L = K(\theta)$, there are number rings $\mathcal{O}_L$ which are not of the form $\mathbb{Z}[\theta]$ for some algebraic integer; for example the number ring of $\mathbb{Q}(\sqrt{7}, \sqrt{10})$ [MS77, Exercise 2.30] can not be written as $\mathbb{Z}[\theta]$ for any algebraic integer θ. Therefore if we want a general purpose prime factorization theorem for extending primes of $\mathcal{O}_K$ to $\mathcal{O}_L$, $L = K(\theta)$, we need

to understand the relationship between $\mathcal{O}_K[\theta]$ and $\mathcal{O}_L$, as well as how this relationship affects extension of ideals of $\mathcal{O}_K[\theta]$ to $\mathcal{O}_L$.

2 Orders and Conductors

An order is a Krull dimension one Noetherian domain $\mathcal{O}$. The Krull-Akizuki theorem [Neu13, Proposition 2.8] says that the integral closure of an order $\mathcal{O}$ in its field of fractions is a dimension one Noetherian ring, hence the integral closure of an order is a Dedekind domain. We are not concerned with the full generality of orders, we are only concerned with the case of orders of number fields so we will only work with the theory in that context.

Definition 1. (Orders)
Let K be a number field with ring of integers $\mathcal{O}_K$. An **order** of K (or $\mathcal{O}_K$) is a Krull dimension one Noetherian subring $\mathcal{O} \subset \mathcal{O}_K$ such that $\mathrm{Frac}(\mathcal{O}) = K$. Equivalently, an order of $\mathcal{O}$ is a subring $\mathcal{O} \subset K$ which is also a finitely generated $\mathbb{Z}$-module of rank $[K : \mathbb{Q}]$.

Orders of a number field K are subrings of $\mathcal{O}_K$ which satisfy all the properties to be a Dedekind domain except for integral closure. The lack of integral closure causes orders $\mathcal{O} \subsetneq \mathcal{O}_K$ to lack the unique factorization property that Dedekind domains have, as well as several other properties that ideals in Dedekind domains have such as invertibility as fractional ideals and two-generation.

If $\mathcal{O} \subset \mathcal{O}_K$ is an order, there is an ideal of $\mathcal{O}$ which determines which primes of $\mathcal{O}$ (and subsequently which ideals) satisfy the properties of ideals in Dedekind domains or if they fail these properties. This ideal is the conductor, and can be characterized in many ways; perhaps the simplest is the conductor of the ring extension $\mathcal{O} \subset \mathcal{O}_K$ is the greatest ideal $\mathfrak{c}$ of $\mathcal{O}$ such that $\mathfrak{c}\mathcal{O}_K = \mathfrak{c}$, and such an ideal exists because $0\mathcal{O}$ is an ideal of $\mathcal{O}_K$ and $\mathcal{O}$ is Noetherian.

Definition 2. (The conductor of an order)
Given an order $\mathcal{O} \subset \mathcal{O}_K$ of a number field K, the **conductor** of $\mathcal{O} \subset \mathcal{O}_K$ (or the conductor of $\mathcal{O}$) is the largest $\mathcal{O}$-ideal which is also an $\mathcal{O}_K$-ideal. That is, the conductor of $\mathcal{O} \subset \mathcal{O}_K$ is the largest ideal $\mathfrak{c} \subset \mathcal{O}$ such that $\mathfrak{c}\mathcal{O}_K \subset \mathfrak{c}$.

Fix an order $\mathcal{O} \subset \mathcal{O}_K$ with conductor $\mathfrak{c}$. If we set $\mathfrak{c}' = \{x \in \mathcal{O}_K \mid x\mathcal{O}_K \subset \mathcal{O}\}$ then it is easy to check that $\mathfrak{c}'$ is an ideal of $\mathcal{O}_K$. If $x \in \mathfrak{c}'$ then $x = x \cdot 1 \in x\mathcal{O}_K \subset \mathcal{O}$ so $\mathfrak{c}' \subset \mathcal{O}$ is also an $\mathcal{O}$-ideal. As $\mathfrak{c}$ is the largest $\mathcal{O}$-ideal which is also an $\mathcal{O}_K$-ideal we have $\mathfrak{c}' \subset \mathfrak{c}$. Conversely, because $\mathfrak{c}\mathcal{O}_K \subset \mathcal{O}$ by $\mathfrak{c}$ being an $\mathcal{O}_K$-ideal contained in $\mathcal{O}$ we have $\mathfrak{c} \subset \mathfrak{c}'$. So

$$\mathfrak{c} = \{x \in \mathcal{O}_K \mid x\mathcal{O}_K \subset \mathcal{O}\}$$

is the conductor of $\mathcal{O} \subset \mathcal{O}_K$ which is also the largest $\mathcal{O}_K$-ideal contained in $\mathcal{O}$. This gives an interpretation of the conductor $\mathcal{O} \subset \mathcal{O}_K$ as being the largest

ideal of $\mathcal{O}$ for which $\mathfrak{c}\mathcal{O}_K = \mathfrak{c}$ (that is, the largest ideal of $\mathcal{O}$ which equals its extension to $\mathcal{O}_K$).

Example 1. Set $K = \mathbb{Q}(i)$ where $i^2 = -1$. Then $\mathcal{O}_K = \mathbb{Z}[i]$. Let $\mathcal{O} = \mathbb{Z} \oplus 2i\mathbb{Z}$. Certainly $\mathcal{O}$ is a subring of $\mathcal{O}_K$ (in particular it contains 1) and is also a free $\mathbb{Z}$-module of rank 2, so indeed $\mathcal{O}$ is an order of K. Note that $2\mathcal{O}_K$ is an ideal of $\mathcal{O}_K$ and $2\mathcal{O}_K = 2\mathbb{Z} \oplus 2i\mathbb{Z} \subset \mathcal{O}$ is an ideal of $\mathcal{O}$. If $\mathfrak{a} \subset \mathcal{O}$ is an ideal of $\mathcal{O}$ and $\mathfrak{a} \subset \mathcal{O}_K$ is an ideal of $\mathcal{O}_K$ then for all $x + iy \in \mathfrak{a}$ both $x + iy$ and $y - ix$ are in $\mathcal{O}$ so both x, y are even hence $\mathfrak{a} \subset 2\mathcal{O}_K$. Thus, the conductor of $\mathcal{O}$ in $\mathcal{O}_K$ is $2\mathcal{O}_K$.

A common situation of interest in the study of elliptic curves are imaginary quadratic orders:

Example 2. Let $K = \mathbb{Q}(\sqrt{-m})$ for $m \in \mathbb{Z}$ squarefree with $-m \equiv 2, 3 \pmod 4$ and $\mathcal{O} = \mathbb{Z} \oplus c\sqrt{-m}\mathbb{Z}$ where $c \in \mathbb{Z}_{>0}$. Then $\mathcal{O}$ is an order of $\mathcal{O}_K = \mathbb{Z}[\sqrt{-m}]$ with conductor $c\mathbb{Z}[\sqrt{-m}]$. If $-m \equiv 3 \pmod 4$ then the order $\mathbb{Z} \oplus c\sqrt{-m}\mathbb{Z}$ has conductor $c\mathcal{O}_K$ if c is even and conductor $2c\mathcal{O}_K$ if c is odd.

A generic example:

Example 3. Let $\mathcal{O}_K$ be any number ring and fix a positive non-unit integer c. Then $\mathcal{O}_K + c\mathcal{O}_K$ is an order in $\mathcal{O}_K$ with conductor $c\mathcal{O}_K$.

Algebraically, one could view the conductor of an order $\mathcal{O} \subset \mathcal{O}_K$ as the ideal of $\mathcal{O}$ which dictates its "bad behavior" relative to $\mathcal{O}_K$. In particular, if $\mathfrak{a} \subset \mathcal{O}$ is an ideal of $\mathcal{O}$ coprime to the conductor $\mathfrak{c}$ then $\mathfrak{a}\mathcal{O}_K$ satisfies all the properties of a nonzero ideal in a Dedekind domain: invertibility as a fractional ideal, arbitrary two-generation, containment implies division, and unique factorization into a product of prime ideals coprime to the conductor. Conversely, any ideal $\mathfrak{a} \subset \mathcal{O}$ with $\mathfrak{a} + \mathfrak{c} \neq \mathcal{O}$, that is $\mathfrak{a}$ is not coprime to the conductor, fails to satisfy the special properties of ideals in Dedekind domains listed prior.

Geometrically, $\mathcal{O}_K$ is a Dedekind domain so $\mathrm{Spec}(O_K)$ is a smooth curve. As $\mathcal{O}$ is not integrally closed $\mathrm{Spec}(\mathcal{O})$ has singular points. A prime $\mathfrak{p} \subset \mathcal{O}$ is a non-singular point of $\mathrm{Spec}(\mathcal{O})$ provided $\mathcal{O}_\mathfrak{p}$ is a DVR (Discrete Valuation ring, equivalently a Dedekind domain with only one maximal ideal), so a prime $\mathfrak{p} \subset \mathcal{O}$ is a singular point if $\mathcal{O}_\mathfrak{p}$ is NOT a DVR. The points of $\mathrm{Spec}(\mathcal{O})$ which are singular are precisely the primes of $\mathcal{O}$ which divide the conductor, which can be seen from the proposition below. Putting it all together: the singular locus of $\mathrm{Spec}(\mathcal{O})$ is the vanishing locus of the conductor of $\mathcal{O}$, and the conductor of $\mathcal{O}$ is the largest ideal of $\mathcal{O}$ whose vanishing locus is the singular locus of $\mathrm{Spec}(\mathcal{O})$.

We will only prove what we need to prove the DK theorem, for more information one can see: Lecture Notes 6 and Problem Set 3 of Drew Sutherland's Algebraic Number Theory course [Sut], Keith Conrad's expository document on conductors [Con, Conductor Ideal of an Order], or Neukirch [Neu13, Chapter I.12].

The following lemma is a basic exercise in commutative algebra.

Lemma 1. *Let K be a number field and $\mathcal{O} \subset K$ an order of K with conductor $\mathfrak{c}$. For any prime $\mathfrak{p} \subset \mathcal{O}$ the following are equivalent:*

1. *$\mathfrak{p}$ does not contain the conductor of $\mathcal{O}$.*
2. *$\mathfrak{p}$ is coprime to the conductor of $\mathcal{O}$.*

Proof. Let $\mathfrak{c}$ be the conductor of $\mathcal{O} \subset \mathcal{O}_K$.

($1 \to 3$): As $\mathfrak{p}$ does not contain $\mathfrak{c}$ the ideal $\mathfrak{p} + \mathfrak{c}$ strictly contains the ideal $\mathfrak{p}$. As $\mathcal{O}$ is dimension one and $\mathfrak{p}$ is nonzero prime ideal of $\mathcal{O}$ we have that $\mathfrak{p}$ is maximal hence $\mathfrak{p} + \mathfrak{c} = \mathcal{O}$.

($3 \to 1$): Suppose $\mathfrak{p} + \mathfrak{c} = \mathcal{O}$. If $\mathfrak{p}$ contained $\mathfrak{c}$ then by maximality of $\mathfrak{p}$ we would have $\mathfrak{p} + \mathfrak{c} = \mathfrak{p} \subsetneq \mathcal{O}$, but this is not the case hence $\mathfrak{p}$ must not contain $\mathfrak{c}$.

□

An important property of primes not containing the conductor is the following corollary:

Corollary 1. *Let K be a number field and $\mathcal{O} \subset \mathcal{O}_K$ an order of K with conductor $\mathfrak{c}$. If $\mathfrak{p}$ is a prime of $\mathcal{O}$ satisfying any of the 3 equivalent conditions in 1 then $(\mathcal{O}_K/\mathfrak{p}\mathcal{O}_K) \cong \mathcal{O}/\mathfrak{p}$ and consequently $\mathfrak{p}\mathcal{O}_K$ is a prime of $\mathcal{O}_K$.*

Proof. We have a homomorphism $\psi : \mathcal{O}/\mathfrak{p} \to \mathcal{O}_K/\mathfrak{p}\mathcal{O}_K$ given by $a+\mathfrak{p} \mapsto a+\mathfrak{p}\mathcal{O}_K$ because $\mathfrak{p}$ is contained in the kernel of the composition $\mathcal{O} \hookrightarrow \mathcal{O}_K \twoheadrightarrow \mathcal{O}_K/\mathfrak{p}\mathcal{O}_K$. As $\mathfrak{p} \subset \mathcal{O}$ is a prime $\mathcal{O}/\mathfrak{p}$ is a field, so the map ψ is either the zero map or an injection. We will prove that ψ is injective and then we will show that ψ is also surjective, this then shows that $\mathcal{O}_K/\mathfrak{p}\mathcal{O}_K \cong \mathcal{O}/\mathfrak{p}$ which shows that $\mathfrak{p}\mathcal{O}_K$ is a maximal ideal as $\mathcal{O}_K/\mathfrak{p}\mathcal{O}_K \cong \mathcal{O}/\mathfrak{p}$ is a field.

As $\mathfrak{p} + \mathfrak{c} = \mathcal{O}$ there exists $x \in \mathfrak{p}$ and $y \in \mathfrak{c}$ such that $1 = x + y$ with $y \notin \mathfrak{p}$. Then $\psi(x + y + \mathfrak{p}) = \psi(1 + \mathfrak{p}) = \psi(y + \mathfrak{p}) = y + \mathfrak{p}\mathcal{O}_K$, and as $y \notin \mathfrak{p}\mathcal{O}_K$ (because $\mathfrak{c} + \mathfrak{p}\mathcal{O}_K = \mathcal{O}_K$) we have $\psi(y + \mathfrak{p}) \in \mathcal{O}_K/\mathfrak{p}\mathcal{O}_K$ is nonzero. So indeed, ψ is injective. Choose any $x + \mathfrak{p}\mathcal{O}_K$ in $\mathfrak{p}\mathcal{O}_K$. As $\mathfrak{c} + \mathfrak{p}\mathcal{O}_K = \mathcal{O}_K$ we can write $x = y + z$ for some $y \in \mathfrak{p}\mathcal{O}_K$ and $z \in \mathfrak{c}$. As $\mathfrak{c} \subset \mathcal{O}$ we have $z \subset \mathcal{O}$ and moreover $\psi(z + \mathfrak{p}) = z + \mathfrak{p}\mathcal{O}_K = x - y + \mathfrak{p}\mathcal{O}_K = x + \mathfrak{p}\mathcal{O}_K$. So then ψ is both injective and surjective, and as noted earlier this completes the proof. □

Why do primes coprime to the conductor matter? If $\mathfrak{p}$ is coprime to the conductor $\mathfrak{c}$ of $\mathcal{O} \subset \mathcal{O}_K$ then $\mathfrak{p}\mathcal{O}_K$ is a prime of $\mathcal{O}_K$ and furthermore $\mathcal{O}/\mathfrak{p} \cong \mathcal{O}_K/\mathfrak{p}\mathcal{O}_K$. This means that primes of $\mathcal{O}$ away from the conductor are "almost" primes of Dedekind domains, they satisfy all properties except actually being inside of one!

Remark 1. In the list of equivalent conditions in 1 one can include: "$\mathfrak{p}$ is an invertible fractional $\mathcal{O}$-ideal", "$\mathcal{O}_\mathfrak{p}$ is a DVR", and "$\mathfrak{p}\mathcal{O}_\mathfrak{p}$ is principal". Proofs of these extra facts can be found in [Con, Conductor Ideal of an Order] or worked out by the reader, but due to their local nature we shall leave them alone because Marcus [MS77] avoids local methods.

This lemma is not needed for the proof of the theorem, but it will connect the historical theorem statement with our perspective of orders and conductors. In particular, we will prove the DK theorem from reasoning involving orders and conductors, but such reasoning is more "modern" compared to the approach of considering prime divisors of the index $|\mathcal{O}_L/\mathcal{O}_K[\theta]|$ as Dedekind and Kummer would have.

Lemma 2. *Fix number fields $K \subset L = K(\theta)$ where θ is an algebraic integer, and fix a prime $\mathfrak{p} \subset \mathcal{O}_K$ as well as the prime $p \in \mathbb{Z}$ lying under $\mathfrak{p}$. Then $\mathfrak{p}\mathcal{O}_K[\theta]$ is coprime to the conductor of $\mathcal{O}_K[\theta] \subset \mathcal{O}_L$ if and only if p does not divide the index of additive groups $|\mathcal{O}_L/\mathcal{O}_K[\theta]|$.*

Remark 2. The assumption that $\mathcal{O}_K[\theta]$ is the order of consideration in $\mathcal{O}_L$ is not necessary, the above theorem applies for any order $\mathcal{O} \subset \mathcal{O}_L$ where $\mathfrak{p}[\theta]$ is replaced with a prime of $\mathcal{O}$.

Proof. Let $\mathfrak{c}$ be the conductor of $\mathcal{O}_K[\theta] \subset \mathcal{O}_L$. Denote by $\mathfrak{p}[\theta]$ the extension of $\mathfrak{p}$ to $\mathcal{O}_K[\theta]$. Note that as $\mathfrak{c}$ is both an $\mathcal{O}_K[\theta]$-ideal and an $\mathcal{O}_L$-ideal we have

$$\mathcal{O}_L/\mathcal{O}_K[\theta] \cong (\mathcal{O}_L/\mathfrak{c})/(\mathcal{O}_K[\theta]/\mathfrak{c})$$

By Lagrange's theorem we then have

$$|\mathcal{O}_L/\mathcal{O}_K[\theta]||\mathcal{O}_K[\theta]/\mathfrak{c}| = |\mathcal{O}_L/\mathfrak{c}|$$

($\rightarrow$): Suppose $\mathfrak{p}[\theta]$ is coprime to the conductor, so $\mathfrak{p}[\theta]+\mathfrak{c} = \mathcal{O}_K[\theta]$ and $\mathfrak{p}\mathcal{O}_L+\mathfrak{c} = \mathcal{O}_L$. Suppose for the sake of contradiction that p divides $|\mathcal{O}_L/\mathcal{O}_K[\theta]|$. By the equality of indices at the beginning of the proof this implies that p divides the index $|\mathcal{O}_L/\mathfrak{c}|$. As $\mathcal{O}_L$ is a number ring we can write $\mathfrak{c} = \prod_{i=1}^r \mathfrak{q}_i^{e_i}$ where each $\mathfrak{q}_i \subset \mathcal{O}_L$ is a prime $\mathcal{O}_L$-ideal. By the CRT we can then write

$$\mathcal{O}_L/\mathfrak{c} \cong \prod_{i=1}^r \mathcal{O}_L/\mathfrak{q}_i^{e_i}$$

and each factor $\mathcal{O}_L/\mathfrak{q}_i^{e_i}$ is a finite ring with isomorphism type $\mathbb{Z}/q_i^m\mathbb{Z}$ where q_i is the prime of $\mathbb{Z}$ lying below $\mathfrak{q}_i$. As p divides the order of $\mathcal{O}_L/\mathfrak{c}$ one of the factors in the product decomposition of $\mathcal{O}_L/\mathfrak{c}$ is $\mathbb{Z}/p^m\mathbb{Z}$ for some integer m. In particular there is a prime $\mathfrak{P} \subset \mathcal{O}_L$ such that $\mathfrak{P}$ lies over p and $\mathfrak{P}$ also divides $\mathfrak{c}$ in $\mathcal{O}_L$. But then $\mathfrak{P}$ must also lie over $\mathfrak{p} \subset \mathcal{O}_K$ hence $\mathfrak{P} \cap \mathcal{O}_K[\theta]$ is a prime ideal of $\mathcal{O}_K[\theta]$ containing $\mathfrak{p}[\theta]$. As $\mathfrak{P}$ is a prime containing $\mathfrak{c}$ and $\mathfrak{p}\mathcal{O}_L$ we have $\mathfrak{P} \supset \mathfrak{P} + \mathfrak{c} \supset \mathfrak{p}\mathcal{O}_L + \mathfrak{c} = \mathcal{O}_L$, which contradicts $\mathfrak{P}$ being a prime ideal of $\mathcal{O}_L$. Thus we conclude p indeed does not divide the index $|\mathcal{O}_L/\mathcal{O}_K[\theta]|$.

($\leftarrow$): Suppose that p does not divide the index $|\mathcal{O}_L/\mathcal{O}_K[\theta]|$. Suppose for the sake of contradiction that $\mathfrak{p}$ is not coprime to $\mathfrak{c}$, and by 1 this means that $\mathfrak{p}[\theta] \supset \mathfrak{c}$. Then $\mathfrak{p}\mathcal{O}_L \supset \mathfrak{c}$ hence there is a prime $\mathfrak{P} \subset \mathcal{O}_L$ such that $\mathfrak{P} \supset \mathfrak{p}\mathcal{O}_L \supset \mathfrak{c}$ hence $\langle p \rangle = \mathfrak{P} \cap \mathbb{Z} \supset \mathfrak{c} \cap \mathbb{Z}$. As in the prior direction we can then conclude $\mathcal{O}_L/\mathfrak{c}$ has a subgroup isomorphic to $\mathbb{Z}/p^m\mathbb{Z}$ for some integer m and hence p divides $|\mathcal{O}_L/\mathfrak{c}|$. As $|\mathcal{O}_L/\mathfrak{c}| = |\mathcal{O}_L/\mathcal{O}_K[\theta]||\mathcal{O}_K[\theta]/\mathfrak{c}|$ we have that p divides the product $|\mathcal{O}_L/\mathcal{O}_K[\theta]||\mathcal{O}_K[\theta]/\mathfrak{c}|$ and by assumption that p does not divide the index $|\mathcal{O}_L/\mathcal{O}_K[\theta]|$ we conclude that p divides $|\mathcal{O}_K[\theta]/\mathfrak{c}|$. By Cauchy's theorem we can then produce an $x + \mathfrak{c} \in \mathcal{O}_K[\theta]/\mathfrak{c}$ such that $px \equiv 0 \pmod{\mathfrak{c}}$ and $x \notin \mathfrak{c}$. But then $x \in \mathcal{O}_L$ as $\mathcal{O}_K[\theta] \subset \mathcal{O}_L$ and $x \notin \mathfrak{c}$, so $\mathfrak{p}x \equiv 0 \pmod{()\mathfrak{c}}$ in $\mathcal{O}_L/\mathfrak{c}$ as well. Therefore $\mathcal{O}_L/\mathfrak{c}$ has an element of order p which implies that p divides $|\mathcal{O}_L/\mathfrak{c}|$, contradicting our intial assumption. So then indeed, $\mathfrak{p}[\theta] + \mathfrak{c} = \mathcal{O}_K[\theta]$. $\square$

3 The Main Theorem

Theorem 1. *(Dedekind-Kummer Index Theorem)*
[MS77, Thm 27] [Neu13, Proposition I.8.3]
Fix number fields $K \subset L = K(\theta)$ where θ is an algebraic integer, and fix a prime $\mathfrak{p} \subset \mathcal{O}_K$ satisfying either of the equivalent conditions

- *The prime $p \in \mathbb{Z}$ lying below $\mathfrak{p} \subset \mathcal{O}_K$ does not divide the index of additive groups $|\mathcal{O}_L/\mathcal{O}_K[\theta]|$.*
- *$\mathfrak{p}\mathcal{O}_K[\theta]$ is coprime to the conductor of $\mathfrak{p}\mathcal{O}_K[\theta] \subset \mathcal{O}_L$.*

Let $g \in \mathcal{O}_K[x]$ be the minimal polynomial for θ over K and let $\overline{g}$ denote g reduced to $(\mathcal{O}_K/\mathfrak{p})[x]$. Factor $\overline{g}$ in $\mathbb{F}_\mathfrak{p}[x]$ into distinct prime power factors as $\overline{g} = \prod_{i=1}^r \overline{g}_i^{e_i}$ and for each $1 \leq i \leq r$ let $g_i \in \mathcal{O}_K[x]$ be a lift of $\overline{g}_i$ to $\mathcal{O}_K[x]$. For each $1 \leq i \leq r$ set

$$\mathfrak{q}_i = \mathfrak{p}\mathcal{O}_L + g_i(\theta)\mathcal{O}_L = \langle \mathfrak{p}, g_i(\theta) \rangle \subset \mathcal{O}_L$$

Then

1. *Each $\mathfrak{q}_i$ is a prime of $\mathcal{O}_L$.*
2. *$\mathfrak{p}\mathcal{O}_L = \prod_{i=1}^r \mathfrak{q}_i^{e_i}$. Consequently each $\mathfrak{q}_i$ lies over $\mathfrak{p}$ with ramification degree $e(\mathfrak{q}_i|\mathfrak{p}) = e_i$.*
3. *Each $\mathfrak{q}_i$ lies over $\mathfrak{p}$ with inertial degree $f(\mathfrak{q}_i|\mathfrak{p}) = \mathrm{Deg}(g_i)$.*

The proof of [MS77, Theorem 27] uses the second condition (which follows the historical statement of the theorem), and the proof of Neukirch [Neu13, I.8.3] uses the first condition. We will combine the proofs of Marcus and Neukirch by following Marcus [MS77] first and then use the ideas presented in [Neu13] to finalize the proof.

Proof. $\mathfrak{p}$ is a prime of $\mathcal{O}_K$ so $1 \notin \mathfrak{p}$ so $\mathrm{Deg}(g_i) = \mathrm{Deg}(\overline{g}_i)$ for all $1 \leq i \leq r$. For each $1 \leq i \leq r$ let $f_i = \mathrm{Deg}(g_i)$. First suppose each of the following assumptions are true:

a) For each $1 \leq i \leq r$ the quotient $\mathcal{O}_L/\mathfrak{q}_i$ is a field of order $\#\left(\mathcal{O}_K/\mathfrak{p}\right)^{f_i}$.

b) For $1 \leq i \neq j \leq r$ we have $\mathfrak{q}_i + \mathfrak{q}_j = \mathcal{O}_L$.

c) $\mathfrak{p}\mathcal{O}_L$ divides $\prod_{i=1}^r \mathfrak{q}_i^{e_i}$.

Assumption (a) and the definition of the $\mathfrak{q}_i$ prove conclusion 3, assumption (a) also proves item 1 as each $\mathcal{O}_L/\mathfrak{q}_i$ is a field. Assumption (b) shows that the $\mathfrak{q}_i$ are all distinct. As $\mathfrak{p}\mathcal{O}_L$ divides $\prod_{i=1}^r \mathfrak{q}_i^{e_i}$ and all the $\mathfrak{q}_i$ are distinct we conclude that $\{\mathfrak{q}_1, \ldots, \mathfrak{q}_r\}$ are precisely the primes of $\mathcal{O}_L$ lying over $\mathfrak{p}$ each of ramification degree $e(\mathfrak{q}_i|\mathfrak{p}) = d_i \leq e_i$. By [MS77, Theorem 21] if we set $n = [L : K]$ then

$$n = \sum_{i=1}^r d_i f_i \leq \sum_{i=1}^r e_i f_i$$

As g is the minimal polynomial for θ over $\mathcal{O}_K$ and the extension degree of $L = K(\theta) \supset K$ is n we conclude that g has degree n. As $\mathrm{Deg}(g) = \mathrm{Deg}(\overline{g})$ and we factored

$$g \equiv g_1^{e_1} \ldots g_r^{e_r} \pmod{\mathfrak{p}}$$

we have $n = \mathrm{Deg}(g) = \sum_{i=1}^r e_i \,\mathrm{Deg}(g_i)$, and by assumption $f_i = \mathrm{Deg}(g_i)$ for each $1 \leq i \leq r$. Then

$$n = \sum_{i=1}^r d_i f_i \leq \sum_{i=1}^r e_i f_i = \sum_{i=1}^r \sum_{i=1}^r e_i \,\mathrm{Deg}(g_i) = n$$

hence $\sum_{i=1}^r d_i f_i = \sum_{i=1}^r e_i f_i$ with each $d_i \leq e_i$. As all $d_i, e_i, f_i \geq 1$ it follows by a basic bounding argument that if any $d_i < e_i$ the aforementioned equality would fail to hold, hence we conclude $d_i = e_i$ for all $1 \leq i \leq r$.

Now we must show that assumptions a,b,c hold.

(a): We can extend $\mathfrak{p} \subset \mathcal{O}_K$ to $\mathfrak{p}[x] \subset \mathcal{O}_K[x]$ or $\mathfrak{p}[\theta] \subset \mathcal{O}_K[\theta]$. As g is the minimal polynomial of θ over $\mathcal{O}_K$ we have

$$\mathcal{O}_K[\theta] \cong \mathcal{O}_K[x]/\langle g \rangle$$

For each $1 \leq i \leq r$ set $\hat{\mathfrak{q}}_i = \mathfrak{p}[\theta] + \langle g_i(\theta) \rangle$. We have

$$\mathcal{O}_K[\theta]/\hat{\mathfrak{q}}_i = (\mathcal{O}_K[x]/\langle g \rangle)/\left((\mathfrak{p}[x] + g_i(x))/(\langle g \rangle)\right) = \mathcal{O}_K[x]/(\mathfrak{p}[x] + g_i) \cong (\mathcal{O}_K/\mathfrak{p})[x]/\langle \overline{g}_i \rangle$$

As $\overline{g}_i \in (\mathcal{O}_K/\mathfrak{p})[x]$ is irreducible of degree f_i the quotient $(\mathcal{O}_K/\mathfrak{p})[x]/\langle \overline{g} \rangle$ is a field extension of $\mathcal{O}_K/\mathfrak{p}$ with degree $f_i = \mathrm{Deg}(g_i)$, as $\mathcal{O}_K/\mathfrak{p}$ is a finite field this shows that $\mathcal{O}_K[\theta]/\hat{\mathfrak{q}}_i$ is a finite field of order $\#(\mathcal{O}_K/\mathfrak{p})^{f_i}$. So indeed, $\hat{\mathfrak{q}}_i$ is a prime ideal of $\mathcal{O}_K[\theta]$. As $\mathfrak{p}$ does not divide the conductor $\mathfrak{c}$ of $\mathcal{O}_K[\theta]$ we

conclude as well that $\mathfrak{p}[\theta]$ does not divide $\mathfrak{c}$. Thus we can conclude that $\mathfrak{p}[\theta]$ is coprime to $\mathfrak{c}$ as $\mathfrak{p}[\theta] + \mathfrak{c} \supset \mathfrak{p} + \mathfrak{c} \ni 1$ and furthermore $\hat{\mathfrak{q}}_i$ is coprime to $\mathfrak{c}$ because $\hat{\mathfrak{q}}_i + \mathfrak{c} \supset \mathfrak{p}[\theta] + \mathfrak{c} = \mathcal{O}_K[\theta]$. Then by 1 we know that $\hat{\mathfrak{q}}_i\mathcal{O}_L$ is a prime of $\mathcal{O}_L$, that $\mathfrak{q}_i = \hat{\mathfrak{q}}_i\mathcal{O}_L$, and $\mathcal{O}_L/\mathfrak{q}_i \cong \mathcal{O}_K[\theta]/\hat{\mathfrak{q}}_i$ is a field of order $\#(\mathcal{O}_K/\mathfrak{p})^{f_i}$.

(b): We have $\mathfrak{q}_i + \mathfrak{q}_j = \mathfrak{p}\mathcal{O}_L + \langle g_i(\theta)\rangle + \langle g_j(\theta)\rangle$. In the Euclidean domain $(\mathcal{O}_K/\mathfrak{p})[x]$, the polynomials $\overline{g}_i$ and $\overline{g}_j$ are non-associate irreducible polynomials so we can lift the Bèzout equation in $(\mathcal{O}_K/\mathfrak{p})[x]$ to $\mathcal{O}_K[\theta]$ and produce $x \in \mathfrak{p}$ as well as $a, b \in \mathcal{O}_K$ such that $ag_i(\theta) + bg_j(\theta) = 1 + x$. The element $1 = ag_i(\theta) + bg_j(\theta) - x$ is in $\mathfrak{q}_i + \mathfrak{q}_j \subset \mathcal{O}_L$ which completes the proof $\mathfrak{q}_i, \mathfrak{q}_j$ are comaximal.

(c): Computing directly

$$\mathfrak{a} = \prod_{i=1}^{r}(\mathfrak{p}\mathcal{O}_L + \langle g_i(\theta)\rangle)^{e_i} = (\mathfrak{p}\mathcal{O}_L)^{e_1+\ldots+e_r} + \ldots + \left\langle \prod_{i=1}^{r} g_i(\theta)^{e_i} \right\rangle$$

where the $\ldots$ conceals ideals of the form $(\mathfrak{p}\mathcal{O}_L)^m\langle g_1(\theta)\rangle^{m_1}\ldots\langle g_r(\theta)\rangle^{m_r}$ where each of $m, m_1, \ldots, m_r$ is an integer greater than 0, in particular $\mathfrak{p}$ divides each term of the $\ldots$. As $g(\theta) = 0$ and $g \equiv \prod_{i=1}^{r}\overline{g}_i^{e_i} \pmod{\mathfrak{p}}$ we have that $\prod_{i=1}^{r} g_i(\theta)^{e_i} \in \mathfrak{p}[\theta] \subset \mathfrak{p}\mathcal{O}_L$ hence $\left\langle \prod_{i=1} g_i(\theta)^{e_i} \right\rangle \subset \mathfrak{p}\mathcal{O}_L$. Therefore $\mathfrak{p}$ divides every term in the sum defining $\mathfrak{a}$, which shows that $\mathfrak{p}\mathcal{O}_L$ divides $\mathfrak{a} = \prod_{i=1}^{r}\mathfrak{q}_i^{e_i}$. $\square$

Remark 3. We will discuss the relation between the index $|\mathcal{O}_L/\mathcal{O}_K[\theta]|$ and the conductor $\mathfrak{c}$ of $\mathcal{O}_K[\theta] \subset \mathcal{O}_L$, in particular we will show that if $\mathfrak{p} \subset \mathcal{O}_K$ is a prime lying over $p \in \mathbb{Z}$ such that p divides $|\mathcal{O}_L/\mathcal{O}_K[\theta]|$ then $\mathfrak{p}[\theta]$, the extension of $\mathfrak{p}$ to $\mathcal{O}_K[\theta]$, divides the conductor of $\mathcal{O}_K[\theta] \subset \mathcal{O}_L$.

Remark 4. In the case when $\mathcal{O}_K[\theta] = \mathcal{O}_L$, the DK theorem can be used to factor the extension of every prime of $\mathcal{O}_K$ to $\mathcal{O}_L$. This follows because the index $|\mathcal{O}_L/\mathcal{O}_K[\theta]| = |\mathcal{O}_L/\mathcal{O}_L|$ is 1 hence divisible by no primes of $\mathbb{Z}$, equivalently the conductor of $\mathcal{O}_K[\theta] = \mathcal{O}_L \subset \mathcal{O}_L$ is $\mathcal{O}_L$ which is divisible by no primes of $\mathcal{O}_K$.

4 Examples

First we give a plethora of examples for factoring integer primes.

Example 4. (Quadratic Fields Case 1)
Fix $K = \mathbb{Q}(\sqrt{m})$ with m squarefree. As $m \equiv 2, 3 \pmod 4$ we have $\mathcal{O}_K = \mathbb{Z}[\sqrt{m}]$. As $\mathbb{Z}[\sqrt{m}] \subset \mathbb{Z}[\sqrt{m}]$ is actually an equality, the conductor of the extension is 1 and so every prime of $\mathbb{Z}$ avoids the conductor. In particular we may apply the DK theorem for all p.

Fix any prime $p \in \mathbb{Z}$. The minimal polynomial of $\sqrt{m}$ over $\mathbb{Z}$ is $g(x) = x^2 - m$. If p divides m then $x^2 - m \equiv x^2$ and applying the DK theorem gives

$$p\mathbb{Z}[\sqrt{m}] = \langle p, \sqrt{m}\rangle^2$$

If m is a square mod p then $x^2 - m \equiv (x+a)(x-a) \mod p$ where $a \in \mathbb{Z}$ is a square of m mod p. Applying DK gives

$$p\mathbb{Z}[\sqrt{m}] = \langle p, \sqrt{m}+a\rangle\langle p, \sqrt{m}-a\rangle$$

Finally if m is not a square mod p then $x^2 - m$ is irreducible mod p and

$$p\mathbb{Z}[\sqrt{m}] = \langle p, \sqrt{m}^2 - m\rangle p\mathbb{Z}[\sqrt{m}]$$

which implies that $p\mathbb{Z}[\sqrt{m}]$ is prime. Summarizing:

$p \in \mathbb{Z}$	Factorization	Pattern
Divides m	$\langle p, \sqrt{m}\rangle^2$	Totally ramified
$m \equiv a^2$ a nonzero square mod p	$\langle p, \sqrt{m}+a\rangle\langle p, \sqrt{m}-a\rangle$	Splits completely
m not a square mod p	$p\mathbb{Z}[\sqrt{m}]$	Inert

The above example highlights a connection between ideal factorization in $\mathbb{Z}[\sqrt{m}]$ and whether or not m is a square mod p. This connection informs a proof of quadratic reciprocity which can be seen in [MS77, Chapter 4]

Example 5. (Quadratic Fields Case 2)
Fix $K = \mathbb{Q}(\sqrt{m})$ with m squarefree, and now consider the case in which $m \equiv 1 \pmod 4$. First we need to compute the conductor of $\mathbb{Z}[\sqrt{m}]$ in $\mathcal{O}_K = \mathbb{Z}\left[\frac{1+\sqrt{m}}{2}\right]$. Well, $2\mathbb{Z}\left[\frac{1+\sqrt{m}}{2}\right] = 2\mathbb{Z} \oplus \mathbb{Z}(1+\sqrt{m})$ is an ideal of $\mathcal{O}_K$ which is also an ideal of $\mathbb{Z}[\sqrt{m}]$ so is contained in the conductor. If $\mathfrak{a} \subset \mathbb{Z}[\sqrt{m}]$ is an ideal contained in the conductor then $\left(\frac{1+\sqrt{m}}{2}\right)\mathfrak{a} \subset \mathfrak{a}$ (because $\mathfrak{a}$ is an $\mathcal{O}_K$-ideal) which implies that $\mathfrak{a}$ contains an integer divisible by 2 hence $\mathfrak{a} \subset 2\mathcal{O}_K$. This shows that indeed $2\mathcal{O}_K$ is the conductor of the extension $\mathbb{Z}[\sqrt{m}] \subset \mathbb{Z}\left[\frac{1+\sqrt{m}}{2}\right]$. From the prior case and the fact that $2\mathcal{O}_K$ is the conductor of $\mathbb{Z}[\sqrt{m}] \subset \mathcal{O}_K$ we can deduce that whenever $p \in \mathbb{Z}$ is not 2 we have

$$p\mathbb{Z}\left[\frac{1+\sqrt{m}}{2}\right] = \begin{cases} \langle p, \sqrt{m}\rangle^2 & p|m, p \neq 2 \\ \langle p, \sqrt{m}+a\rangle\langle p, \sqrt{m}-a\rangle & m \equiv a^2 \mod p \\ p\mathbb{Z}\left[\frac{1+\sqrt{m}}{2}\right] & m \text{ not a square mod } p \end{cases}$$

Via direct calculations we see

$$2\mathbb{Z}\left[\frac{1+\sqrt{m}}{2}\right] = \begin{cases} \langle 2, \sqrt{m}\rangle^2 & 2|m \\ \left\langle 2, \frac{1+\sqrt{m}}{2}\right\rangle\left\langle 2, \frac{1-\sqrt{m}}{2}\right\rangle & m \equiv 1 \pmod 8 \\ 2\mathbb{Z}\left[\frac{1+\sqrt{m}}{2}\right] & m \equiv 5 \pmod 8 \end{cases}$$

The details are in [MS77, Chapter 3, Theorem 25]. Alternatively one can recall that $x^2 + x + 1$ is irreducible mod 2 and apply the Dedekind-Kummer theorem

to $\mathbb{Z} \subset \mathbb{Z}\left[\frac{1+\sqrt{m}}{2}\right] = \mathcal{O}_K \subset \mathcal{O}_K$ which has minimal polynomial $x^2 - x + \frac{1-m}{4}$ over $\mathbb{Z}$.

The following example is from problem set 2 of Drew Sutherland's ANT course [Sut, Problem Set 2].

Example 6. (Cubic Field)
Suppose $K = \mathbb{Q}(\sqrt[3]{m})$ with m cubefree. Then for all primes $p \in \mathbb{Z}$ we have $p\mathcal{O}_K = \prod_{i=1}^{r} \mathfrak{p}_i^{e_i}$ and because $[\mathbb{Q}(\sqrt[3]{m}) : \mathbb{Q}] = 3$ we have $3 = \sum_{i=1}^{r} e(\mathfrak{p}_i|p)f(\mathfrak{p}_i|p)$. Numerically the possibilities are as follows, where we let $e_i = e(\mathfrak{p}_i|p)$ and $f_i = f(\mathfrak{p}_i|p)$ for each $1 \leq i \leq 3$: Take $K = \mathbb{Q}(\sqrt[3]{5})$. The number ring of K is

r	e_1	e_2	e_3	f_1	f_2	f_3
1	3			1		
1	1			3		
2	1	1		1	2	
2	1	2		1	1	
3	1	1	1	1	1	1

$\mathcal{O}_K = \mathbb{Z}[\sqrt[3]{5}]$. The element $\sqrt[3]{5}$ has minimal polynomial $g(x) = x^3 - 5$. Modulo 2,3,5,7,11, 13 we have Giving the factorizations

p	$g \mod (p)$	Factorization	(e_1, e_2, e_3)	(f_1, f_2, f_3)
2	$x^3 + 1$	$(x+1)(x^2+x+1)$	$(1,1,0)$	$(1,2,0)$
3	$x^3 + 1$	$(x+1)^3$	$(3,0,0)$	$(1,0,0)$
5	x^3	x^3	$(3,0,0)$	$(1,0,0)$
7	$x^3 + 2$	$x^3 + 2$	$(1,0,0)$	$(3,0,0)$
11	$x^3 + 6$	$(x+8)(x^2+3x+9)$	$(1,1,0)$	$(1,2,0)$
13	$x^3 + 8$	$(x+2)(x+5)(x+6)$	$(1,1,0)$	$(1,2,0)$

$$p\mathbb{Z}[\sqrt[3]{5}] = \begin{cases} \langle 2, 1+\sqrt[3]{5}\rangle\langle 2, \sqrt[3]{5^2}+\sqrt[3]{5}+1\rangle & p = 2 \\ \langle 3, 1+\sqrt[3]{5}\rangle^3 & p = 3 \\ \langle 5, \sqrt[3]{5}\rangle^3 & p = 5 \\ 7\mathbb{Z}[\sqrt[3]{5}] & p = 7 \\ \langle 11, \sqrt[3]{5}+8\rangle\langle 11, \sqrt[3]{5^2}+3\sqrt[3]{5}+9\rangle & p = 11 \\ \langle 13, \sqrt[3]{5}+2\rangle\langle 13, \sqrt[3]{5}+5\rangle\langle 13, \sqrt[3]{5}+6\rangle & p = 13 \end{cases}$$

The above factorizations represent every numerically possible case for a factorization of an integer prime in a cubic number ring, except for the case where a prime $p \in \mathbb{Z}$ has ramification degrees $(2,1,0)$ and inertial degrees $(1,1,0)$. The

number ring $\mathcal{O}_K = \mathbb{Z}[\sqrt[3]{5}]$ has integral basis $\{1, \sqrt[3]{5}, \sqrt[3]{5^2}\}$ hence has discriminant

$$\begin{aligned}
\mathrm{Disc}(\mathcal{O}_K) &= \mathrm{Disc}(1, \sqrt[3]{5}, \sqrt[3]{5^2}) \\
&= \begin{vmatrix} \mathrm{Tr}(1\cdot 1) & \mathrm{Tr}(1\cdot \sqrt[3]{5}) & \mathrm{Tr}(1\cdot \sqrt[3]{5^2}) \\ \mathrm{Tr}(\sqrt[3]{5}\cdot 1) & \mathrm{Tr}(\sqrt[3]{5}\cdot \sqrt[3]{5}) & \mathrm{Tr}(\sqrt[3]{5}\cdot \sqrt[3]{5^2}) \\ \mathrm{Tr}(\sqrt[3]{5^2}\cdot 1) & \mathrm{Tr}(\sqrt[3]{5^2}\cdot \sqrt[3]{5}) & \mathrm{Tr}(\sqrt[3]{5^2}\cdot \sqrt[3]{5^2}) \end{vmatrix} \\
&= \begin{vmatrix} \mathrm{Tr}(1) & 0 & 0 \\ 0 & 0 & \mathrm{Tr}(5) \\ 0 & \mathrm{Tr}(5) & 0 \end{vmatrix} \\
&= \begin{vmatrix} 3 & 0 & 0 \\ 0 & 0 & 15 \\ 0 & 15 & 0 \end{vmatrix} \\
&= -3\cdot 15^2 \\
&= -3^3\cdot 5^2
\end{aligned}$$

If any prime other than 3 or 5 ramified in $\mathcal{O}_K$ then it would divide $-27\cdot 15$ which is divisible by only 3 and 5. Therefore the case where a prime p ramifies in $\mathcal{O}_K$ with ramification index 2 is impossible for our choice of K as both the primes 3,5 that ramify in $\mathcal{O}_K$ are totally ramified.

If we replaced K by $K = \mathbb{Q}(\theta)$ where θ has minimal polynomial $x^3 - x - 1$ then $\mathcal{O}_K = \mathbb{Z}[\theta]$. One can see [MS77, Exercise 3.13] that

$$23\mathbb{Z}[\theta] = \langle 23, \theta - 10\rangle^2 \langle 23, \theta - 3\rangle$$

so K is an example of a cubic field with a number ring $\mathcal{O}_K$ such that a prime $p \in \mathbb{Z}$ factors in $\mathcal{O}_K$ with inertial indices $(2, 1, 0)$ and ramification indices $(1, 1, 0)$. One can use the DK theorem to check this.

Example 7. (Prime Cyclotomic Fields)
Fix an odd prime q and the primitive q-th root of unity $\zeta = e^{\frac{2i\pi}{q}}$. The number ring of $K = \mathbb{Q}(\zeta)$ is $\mathcal{O}_K = \mathbb{Z}[\zeta]$ and ζ has minimal polynomial $g(x) = x^{q-1} + \ldots + x + 1$ over $\mathbb{Z}$. Mod q we have

$$g(x) \equiv \frac{x^q - 1}{x - 1} \equiv \frac{(x-1)^q}{x-1} \equiv (x-1)^{q-1} \pmod{q}$$

which gives

$$q\mathbb{Z}[\zeta] = \prod_{j=1}^{q-1} \langle q, \zeta^j - 1\rangle$$

Now suppose p is a prime not equal to q. If $q|(p-1)$ then $\mathbb{F}_p^\times$ has an element of order q by Cauchy's theorem, so there is an $a \in \mathbb{Z}$ such that a is a primitive qth root of unity mod p. Then

$$g(x) \equiv \prod_{j=1}^{q-1} (x - a^j) \pmod{p}$$

and

$$p\mathbb{Z}[\zeta] = \prod_{j=1}^{q-1} \langle p, \zeta - a^j \rangle$$

Suppose instead that $q \nmid (p-1)$. Then there are no qth roots of unity mod p and $g(x)$ is irreducible mod p because $g(x) \pmod p$ is the minimal polynomial for the field extension $\mathbb{F}_p \subset \mathbb{F}_{p^q}$. Therefore $p\mathbb{Z}[\zeta]$ is a prime ideal of $\mathbb{Z}[\zeta]$.

For the purpose of seeing the full generality of the theorem we will give an example with two number rings neither of which is $\mathbb{Z}$.

Example 8. (A Number Ring in a Number Ring)
Let $\zeta = e^{\frac{2i\pi}{6}}$ be a primitive 6th root of unity, and note that ζ^2 is then a primitive third root of unity. Then we have a degree 2 field extension $\mathbb{Q}(\zeta^2) \subset \mathbb{Q}(\zeta)$ of the third cyclotomic field inside the 6th, leading to the extension of number rings $\mathbb{Z}[\zeta^2] \subset \mathbb{Z}[\zeta]$. We know that

$$N_{\mathbb{Q}}^{\mathbb{Q}(\zeta^2)}(1-\zeta^2) = (1-\zeta^2)(1-\zeta^4) = 1 - (\zeta^2 + \zeta^4) + \zeta^6 = 1 - -1 + 1 = 3$$

So the ideal $(1-\zeta^2)\mathbb{Z}[\zeta^2]$ is a prime ideal of $\mathbb{Z}[\zeta^2]$. The minimal polynomial of ζ over $\mathbb{Z}[\zeta^2]$ is $g(x) = x^2 - \zeta^2 \in \mathbb{Z}[\zeta^2][x]$. Factoring g modulo $1-\zeta^2$ we have

$$g(x) \equiv x^2 - 1 \equiv (x+1)(x-1) \mod 1-\zeta^2$$

Applying the DK theorem gives the factorization of $\mathbb{Z}[\zeta]$-ideals

$$(1-\zeta^2)\mathbb{Z}[\zeta] = \langle 1-\zeta^2, 1+\zeta \rangle \langle 1-\zeta^2, 1-\zeta \rangle$$

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